

MASTER EQUATION MODELS FOR CHEMICAL REACTIONS OF IMPORTANCE IN COMBUSTION

Michael J. Pilling¹ and Struan H. Robertson²

¹*Department of Chemistry, University of Leeds, Leeds, LS2 9JT, United Kingdom;
email: M.J.Pilling@chemistry.leeds.ac.uk*

²*Accelrys Inc., Cambridge, CB4 0WN, United Kingdom*

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■ **Abstract** The master equation provides a quantitative description of the interaction between collisional energy transfer and chemical reaction for dissociation, isomerization, and association processes. The approach is outlined for both irreversible and reversible dissociation, isomerization, and association reactions. There is increasing interest, especially in combustion, in association reactions that involve several linked potential wells, with the possibility of isomerization, collisional stabilization, and dissociation along several product channels. A major aim of the application of the master equation to such systems is the linking of the eigenvalues obtained by its solution to the rate coefficients for the phenomenological chemical reactions that describe the system and that are used in combustion models. The approach is illustrated by reference to the reactions $\text{C}_2\text{H}_5 + \text{O}_2$, $\text{H} + \text{SO}_2$, and the dissociation and isomerization of alkyl radicals.

1. INTRODUCTION

Many chemical reactions proceed through the formation of bound intermediates. These intermediates can be stabilized by collision with bath gas or solvent molecules, or they can isomerize or dissociate to form products or, indeed, to regenerate the reactants. In the gas phase, both the rate coefficient and the distribution of products, i.e., of stabilized intermediates and dissociation products, can depend not only on temperature but also on pressure. The dependences can be quite complex, yet these rate- and product-yield data are essential in modelling combustion systems in simulations, for example, of engines. Furthermore, the conditions involved in combustion applications are often exacting, so that direct experimental measurement, at the requisite high pressures and temperatures, can be extremely difficult. Accordingly, it is essential to be able to model the reactions realistically. These models provide not only the basis for the development of a fundamental understanding of these interesting reactions, but also an essential tool for extrapolation of available experimental data and for representing those data in combustion models.

Section 2 of this review discusses the basis of the master equation (ME), which has become a powerful model for pressure-dependent reactions in combustion and other systems, e.g., in atmospheric chemistry and in chemical vapor deposition. Pressure-dependent reactions are typified by unimolecular dissociation and isomerization reactions that form a central component of chemical kinetics (1–4). The reactions depend on the collisional transfer of energy to a reactant to excite it above the dissociation or isomerization threshold. The ME describes this process of collisional transfer between the rovibrational energy states of the reactant and the chemical reaction from those energy states. The collisional energy-transfer process can be modelled as a biased random walk and so can be analyzed using the techniques of stochastic process theory, particularly those related to Markov chains, i.e., to processes for which the probability distribution of transitions to new states of the system depends only on the current state. The ME describes the evolution of these processes (5, 6). The reactive steps correspond to passage over transition states, calling for their representation in terms of transition state theory, necessarily in microcanonical form because of the energy resolved nature of the ME (4). Ab initio calculations of potential energy surfaces are widely employed to characterize both the isomer wells and the transition states (7).

ME calculations were first applied to simple unimolecular dissociation reactions (1–3). The solution of the ME is an expansion in terms of the eigenfunctions and eigenvalues of a matrix operator, which contains the rates of transitions between the energy states and of reaction from those states. In order to limit the dimension of the matrix, to facilitate numerical solution, the energy states are bundled together to form energy “grains,” and the number of eigenpairs, m , is equal to the number of grains. All the eigenvalues are less than or equal to zero. The collisional relaxation of the molecular system evolves temporally as a sum of exponentials with reciprocal time constants equal to the eigenvalues. $m-1$ eigenvalues relate to the process of collisional relaxation, whereas the magnitude of the smallest eigenvalue, $|\lambda_1|$, is equal to $k_{\text{uni}}(p, T)$, the unimolecular dissociation rate coefficient, and the target of the calculations. Similar considerations apply to association and isomerization reactions.

For more complex reactions involving several intermediate energy wells and several product channels, the procedure is somewhat less direct. There are now several eigenvalues associated with the chemical processes. From a combustion perspective, the aim of the calculation is to relate the eigenvalues from the solution of the ME to the phenomenological rate coefficients for the coupled chemical reactions in the scheme defined by the reactants, intermediates, and products. Three examples of such chemical systems are examined in more detail in Section 3 to illustrate some of the principles involved. The first is $\text{C}_2\text{H}_5 + \text{O}_2$, which is a prime example of an important generic reaction occurring in the autoignition of alkanes, which can give rise to knock in spark ignition engines. The reaction illustrates the evolution of the chemical relaxation modes as the temperature is increased and the complexities that occur as two chemical eigenvalues

become comparable in magnitude. It has been carefully investigated from experimental (8–14), potential energy surface (15–18), and ME perspectives, and clearly demonstrates the additional understanding that derives from a ME analysis. The second example involves the reaction $\text{H} + \text{SO}_2$, which is important in understanding the effects of sulfur chemistry in combustion systems. It is slightly more complex than the $\text{C}_2\text{H}_5\text{O}_2$ system and is far less well studied. It illustrates further the relationship between single step chemical processes and chemical relaxation modes. Finally, we look briefly at the isomerization and dissociation of alkyl radicals. Once again this is a system where the nature of the eigenmodes changes as the temperature increases. A key feature of these reactions, though, is the overlap of collisional and chemical relaxation times, which leads to complications in the description of these important reactions in combustion systems.

2. THE MASTER EQUATION

2.1. Energy Transfer

Energy transfer in dilute gas-phase systems occurs as a consequence of inelastic bimolecular collisions. The definition of such a collision—the criteria used to indicate that a pair of molecules have entered into and emerged from a collision—is a complex matter (21). However, for a dilute system, the time between collisions is usually much longer than the time a molecular pair spends in collision. Between collisions, the values of various constants of motion are well defined, and they can be used to specify the state of a molecule, by far the most important being its total internal energy, E . There is a chance that a molecule will emerge from the collision with a different value for E , as energy may have been transferred to or acquired from the collision partner. Over the course of many collisions, E may take many values, and the resulting trajectory is a random walk among the allowed values or state space of E . Thus, these systems can be analyzed using the theory of stochastic processes.

Although quantities such as E remain constant between collisions, the configuration of a molecule evolves in accordance with the laws of motion. At the time of a subsequent collision, there will, in general, be little correlation with the configuration the molecule had on emerging from a previous collision. Thus, to a very good approximation, the energy transfer characteristics will be determined only by the value of E , and any other relevant constants of motion, deriving from the previous collision. Such processes, where the probability of occupying a given state of the system depends only on the immediately previous state and no other, are common in a number of other fields, are referred to as Markov processes, and have been extensively studied (5).

The evolution of a stochastic trajectory is described by the transition rate, which, for the energy transfer system, is written as $W(E|E'; t)$, and in words expresses the probability of transition from a state of energy E' to a state of energy E in time

t . If a process is Markovian then it can be shown (22) that the transition coefficient obeys the Chapman-Kolmogorov equation:

$$W(E|E'; t + t') = \sum_{E''} W(E|E''; t)W(E''|E'; t'). \quad 1.$$

The time intervals t and t' are nonoverlapping and adjacent. Repeated application of Equation 1 provides a description of the stochastic process over any given time interval. It is more useful for our purposes to investigate the properties of Equation 1 as $t \rightarrow 0$. When $t = 0$, $W(E|E''; t)$ must tend to $\delta(E - E'')$ because the system must be in the same state it was in at the end of the time interval t' . Expanding $W(E|E''; t)$ about $t \rightarrow 0$ gives,

$$W(E|E''; t \rightarrow 0) = \delta(E - E'') + tk(E|E'') + o(t), \quad 2.$$

where $o(t)$ represents terms in t^2 or higher. The term $k(E|E'')$ represents the rate of transition between E and E'' and is referred to as the transition coefficient. Substitution of Equation 2 into Equation 1, followed by some rearrangement, gives,

$$\frac{W(E|E'; t + t') - W(E|E'; t')}{t} = \sum_{E''} k(E|E'')W(E''|E'; t'), \quad 3.$$

or, ongoing to the limit $t \rightarrow 0$,

$$\frac{\partial}{\partial t} W(E|E'; t) = \sum_{E''} k(E|E'')W(E''|E'; t). \quad 4.$$

Equation 4 is referred in the mathematical literature as the forward equation and in the physical sciences literature as the ME. It is the equation of motion of the transition coefficient. For the energy transfer system, a slightly more useful form of Equation 4 can be obtained by observing that the probability that a system will be in state E at time t , $\rho(E, t)$, is given by,

$$\rho(E, t) = \sum_{E'} W(E|E'; t)\rho(E', 0). \quad 5.$$

Combining Equations 4 and 5 gives,

$$\frac{\partial}{\partial t} \rho(E, t) = \sum_{E'} k(E|E')\rho(E', t). \quad 6.$$

Equation 6 represents the equation of motion of the probability distribution, or the population density of the system, and it is the solution of this equation that will yield the rate coefficients that we seek.

Before proceeding, it is clear that the function $k(E|E')$ must be defined before the time evolution $\rho(E, t)$ of can be determined. The generality of the ME means that this information must be supplied from elsewhere. In principle, $k(E|E')$ can

be calculated from scattering theory and can be related to the energy dependent cross-section via

$$k_s(E|E') = \left(\frac{8kT}{\pi\mu}\right)^{1/2} \int_0^\infty E_{rel} \sigma(E|E'; E_{rel}) \exp(-\beta E_{rel}) dE_{rel}, \quad 7.$$

where E_{rel} is the relative translational energy of the reactant and colliding bath gas molecule, σ is the cross-section, and the subscript s indicates that k is calculated from scattering theory (23).

There are a number of difficulties in using Equation 7 as it stands. The first problem concerns the normalization properties of $W(E|E'; t)$: in the absence of reaction, density is conserved, and $W(E|E'; t)$ is normalized:

$$\sum_E W(E|E'; t) = 1. \quad 8.$$

The definition of $W(E|E'; t)$ in Equation 2 can only hold if

$$\sum_E k(E|E') = 0, \quad 9.$$

which is simply an alternative way of expressing the conservation of density. However, insertion of Equation 7 into the left-hand side of Equation 9 leads to the definition of the overall collision frequency:

$$\omega(E') = \sum_E k_s(E|E'), \quad 10.$$

which is clearly inconsistent with Equation 9. The resolution of this inconsistency is to define k in terms of k_s as follows:

$$k(E|E') = \begin{cases} k_s(E'|E) & E \neq E' \\ -\sum_{E \neq E'} k_s(E|E') & E = E'. \end{cases} \quad 11.$$

With this definition the form of the ME changes slightly:

$$\frac{\partial}{\partial t} \rho(E, t) = \sum_{E' \neq E} k_s(E|E') \rho(E', t) - \sum_{E' \neq E} k_s(E'|E) \rho(E, t), \quad 12.$$

and using the definition of the collision frequency in Equation 10 gives,

$$\frac{\partial}{\partial t} \rho(E, t) = \sum_{E'} k_s(E|E') \rho(E', t) - \omega(E) \rho(E, t). \quad 13.$$

Having established a connection between the ME and scattering theory there remains the outstanding problem of determining $k_s(E|E')$, and although considerable progress has been made in recent years, experimentally (24, 25) and theoretically (26), the determination of cross-sections for many systems of practical interest remains a challenge. As a result many calculations use heuristic models (2, 3).

The selection of a valid energy transfer model is constrained by two general considerations, one of which is normalization. From Equation 10, a transition probability can be defined as,

$$P(E|E') = \frac{k_s(E|E')}{\sum_E k_s(E|E')} = \frac{k_s(E|E')}{\omega(E')}, \quad 14.$$

and is subject to the normalization condition:

$$\sum_E P(E|E') = 1. \quad 15.$$

Inserting Equation 14 into Equation 13 gives,

$$\frac{\partial}{\partial t} \rho(E, t) = \omega(E) \sum_{E'} P(E|E') \rho(E', t) - \omega(E) \rho(E, t). \quad 16.$$

The other constraint is detailed balance. In the absence of reaction, any system of molecules immersed in a bath gas will tend to an equilibrium distribution, $f(E)$. Hence, detailed balance applies to $k_s(E|E')$:

$$k_s(E'|E) f(E) = k_s(E|E') f(E'), \quad 17.$$

or, in terms of $P(E|E')$,

$$\omega(E') P(E'|E) f(E) = \omega(E) P(E|E') f(E'). \quad 18.$$

Equation 18 is often simplified by making the common approximation that the collision frequency is only weakly dependent on the initial (internal) energy of a molecule and so becomes,

$$P(E'|E) f(E) = P(E|E') f(E'). \quad 19.$$

A number of models have been proposed for $P(E|E')$, and perhaps the most frequently used is the exponential down model, which can be expressed for a deactivating collision as,

$$P(E|E') = A(E) \exp(-\alpha(E' - E)) \quad E' \geq E, \quad 20.$$

where $A(E)$ is the normalization coefficient for energy E , and α is a parameter that governs the average change in energy on collision. This model [which derives from scattering theory (27)] expresses the notion that the transfer of small amounts of energy on collision are more frequent, whereas the transfer of large amounts of energy is less probable. The parameter is, at moderate to high energies, equivalent to $\langle \Delta E \rangle_d^{-1}$, where $\langle \Delta E \rangle_d$ is the average energy transferred in a deactivating collision. For weak colliders, such as the inert gases, $\langle \Delta E \rangle_d$ has a value of $\sim 100 \text{ cm}^{-1}$; strong colliders can have values of $\sim 1000 \text{ cm}^{-1}$ or greater.

Equation 19 specifies the transition probability for a deactivating collision; the corresponding activating probability can be found by applying detailed balance,

$$P(E|E') = A(E') \exp(-\alpha(E - E'))(f(E')/f(E)) \quad E < E'. \quad 21.$$

The normalization coefficient $A(E')$ is determined by substituting Equations 20 and 21 into Equation 15.

Having defined the transition probability, the only remaining quantity that must be specified is the collision frequency ω . As stated previously, the definition of a collision, and therefore the collision frequency, is nontrivial. For the purpose of simulation, however, it is usual to use the Lennard-Jones collision frequency with the collision integral defined by Neufeld et al. (28).

This completes the specification of the energy transfer ME and Equation 16 can be solved and the energy transfer characteristics of a given system investigated. Of more interest in the present context is the impact of energy transfer on reaction.

2.2. Unimolecular Dissociation

In terms of the ME, unimolecular reaction can be thought of as the transfer of population density from one set of states, the reactant, to another set of states, the product(s). Thus reaction can be introduced by including an additional transition rate term into Equation 16,

$$\frac{\partial}{\partial t} \rho(E, t) = \omega \sum_{E'} P(E|E') \rho(E', t) - \omega \rho(E, t) - \sum_{E_p} k(\mathbf{E}_p|E) \rho(E, t), \quad 22.$$

where the vector $\mathbf{E}_p$ specifies the state of the products. Typically, information about the state of the products is not available, or is not of interest, and Equation 22 is written as,

$$\frac{\partial}{\partial t} \rho(E, t) = \omega \sum_{E'} P(E|E') \rho(E', t) - \omega \rho(E, t) - k(E) \rho(E, t). \quad 23.$$

This form of the ME will be the basis of most of the subsequent discussion. Before it can be applied though $k(E)$, the microcanonical rate coefficient, must be specified. This can be achieved in a number of ways, by calculation using either Rice, Ramsperger, Kassel, Marcus (RRKM) theory (29) or the Statistical Adiabatic Channel Model (SACM) (30, 31), from experimental canonical high-pressure rate coefficients via inverse laplace transform theory (32, 33), or from direct experimental measurement (34).

When $k(E)$ has been defined, Equation 23 can be solved for the population density, and once the density is known, the rate coefficient of the system can be determined. For some very simple systems it is possible to solve Equation 23 analytically, but it is generally necessary to use numerical techniques. The state space comprises all the rovibrational (and occasionally electronic) eigenstates that, even for a relatively simple molecule such as methane at moderate energies, can

be vast, leading to a corresponding set of differential equations that is impossible to handle even with the aid of the most powerful machines.

The choice of numerical technique is simplified by noting that the right-hand side of Equation 23 is linear in the density and can be rewritten as

$$\frac{\partial}{\partial t} \rho(E, t) = \hat{M} \rho(E, t), \quad 24.$$

where $\hat{M}$ is a linear operator that combines both energy transfer and reaction terms. One general method for solving such an equation is to apply the separation of variables technique, which, on application of boundary conditions, leads to an expansion in terms of the eigenfunctions and eigenvalues of the operator $\hat{M}$:

$$\rho(E, t) = \sum_i c_i \varphi_i(E) \exp(\lambda_i t) \quad 25.$$

where λ_i is the i th eigenvalue of $\hat{M}$ and φ_i is the corresponding eigenfunction. The expansion coefficients c_i are determined from the initial distribution of reactants. Because of the property of detailed balance, it is possible to show that eigenvalues $\hat{M}$ of must be real and, further, for a meaningful solution they must also all be negative.

Once we have the population density, it possible to determine a number of properties of the system including the overall rate coefficient, which is defined via the phenomenological expression,

$$\frac{d[A]}{dt} = -k_u [A], \quad 26.$$

where $[A]$ is the total concentration of reactant and k_u is the overall unimolecular rate coefficient. The total concentration at any given time in terms of $\rho(E, t)$ is given by

$$[A](t) = [A]_0 \sum_E \rho(E, t). \quad 27.$$

Substitution of Equations 25 and 27 into Equation 26 followed by differentiation gives

$$\sum_E \sum_i \lambda_i c_i \varphi_i(E) \exp(\lambda_i t) = k_u \sum_E \sum_i c_i \varphi_i(E) \exp(\lambda_i t). \quad 28.$$

In general it is found that, if the barrier to reaction is large compared to $k_B T$, then the eigenvalue of smallest magnitude is separated from the other eigenvalues by several orders of magnitude (35). Thus, at long times, that is, at times larger than the time between collisions, it is this eigenvalue that determines the evolution of the system. Denoting the eigenvalue of smallest magnitude as λ_1 , it follows that,

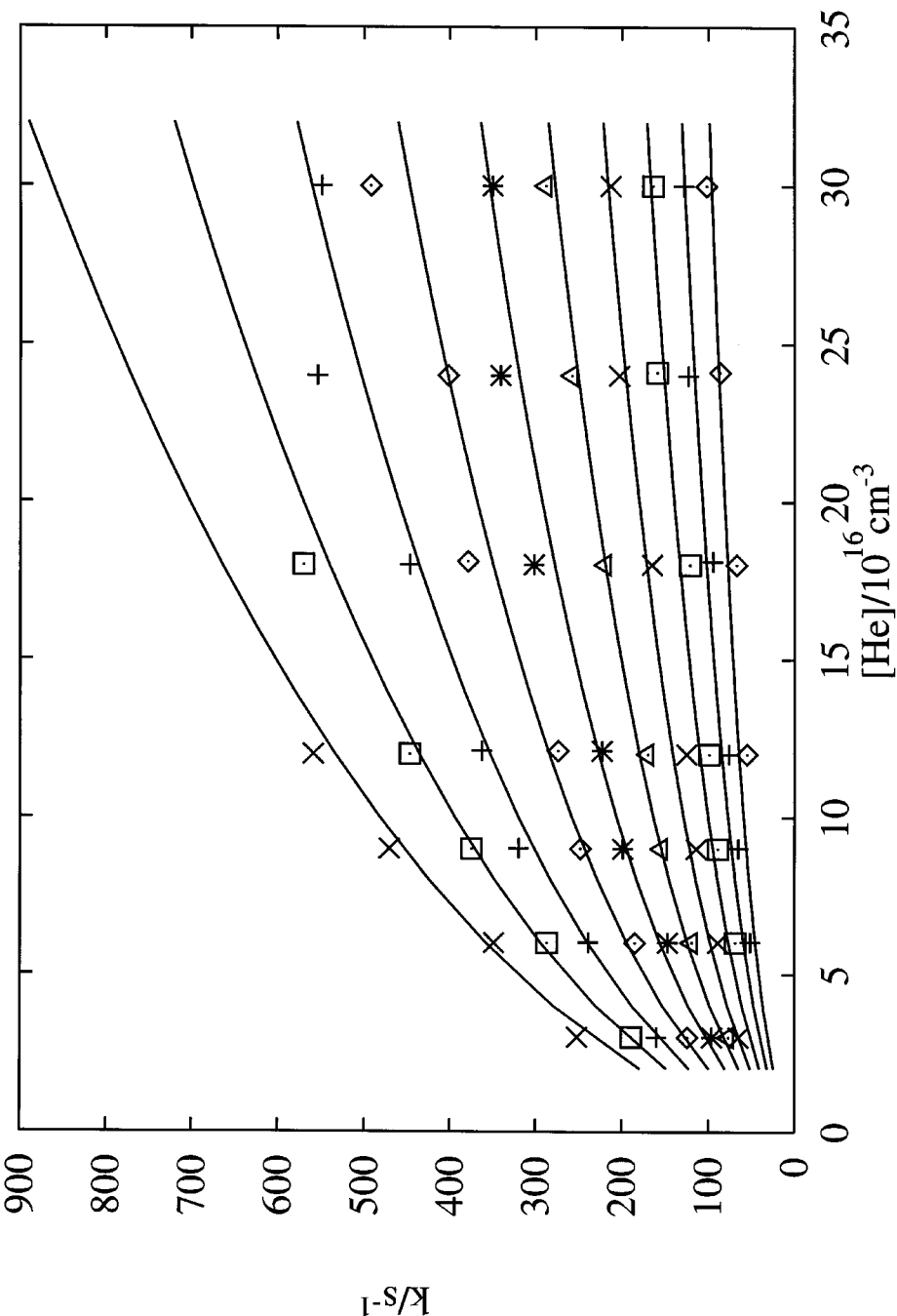
$$k_u = -\lambda_1. \quad 29.$$

So, the determination of the canonical unimolecular rate coefficient amounts to determining the eigenvalue of smallest magnitude of the operator $\hat{M}$. To effect this determination it is necessary to express $\hat{M}$ in a convenient basis set. To date, the most frequently used basis set is one of δ -functions. This choice is referred to as energy graining; the matrix elements are constructed by first dividing the energy axis into a set of contiguous intervals or grains for which mean values of energy, microcanonical rate coefficient and density of states are assigned. The densities of states are usually calculated by a direct-count algorithm, such as that developed by Beyer & Swinehart (3, 36). A δ -function is then used to represent each grain and is located at the mean energy. The complete set of δ -functions forms a basis in which to describe the operator $\hat{M}$. The corresponding equations of motion are usually referred to as the energy grained master equation (EGME).

Figure 1 shows a plot of rate coefficients for the unimolecular dissociation of isopropyl. The points on the plot are experimental values determined using the laser flash photolysis, photoionization mass spectroscopy (37). The lines were obtained by a combined ME/inverse laplace transform approach (38). The inverse laplace transform was used to determine the microcanonical rate coefficients from the limiting high pressure rate coefficient for association ($\text{H} + 2\text{-C}_3\text{H}_7$) (33), using the rate data of Harris & Pitts (39) and the equilibrium constant for the reaction. The model was fitted to the experimental data using a least squares method, with the energy transfer parameter, $\langle \Delta E \rangle_d$, and the enthalpy of formation, $\Delta_f H_{298}^0$, of 2-propyl as the variable parameters, giving $\langle \Delta E \rangle_d = 230 \text{ cm}^{-1}$ (for He as the bath gas) and $\Delta_f H_{298}^0(2\text{-C}_3\text{H}_7) = 88.7 \text{ kJ mol}^{-1}$. This application demonstrates the use of the ME to fit experimental data and to determine molecular parameters—there is a strong sensitivity of the fits to $\Delta_f H_{298}^0(2\text{-C}_3\text{H}_7)$, via the use of the equilibrium constant in the generation of the microcanonical rate constants by inverse laplace transformation of the association-rate data, so that the approach forms a good way of determining enthalpies of formation. Once the system has been characterized in this way, the ME can be used to generate T and p dependent rate data, for combustion or other applications, by simulation. Similarly, if potential energy surface features are known, one can use the ME directly to predict rate constants. Increasingly, ab initio data are being used in this regard (19, 40). Ab initio calculations can also be used to reduce the number of variables in fits to experimental data.

Although the eigen expansion (25), coupled with a δ function basis, is one approach to solving Equation 24, it is not the only approach (41, 42). A technique that is very common in the analysis of stochastic systems is to use Monte Carlo simulation (43–46). The basis of this approach is to repeatedly integrate stochastic trajectories over the timescale of interest and accumulate distributions and averages as the simulation proceeds. There are a number of advantages, not the least of which is it is very economical with respect to computational resources, not having to maintain large matrices.

The above discussion has explicitly assumed that the modes associated with collisional relaxation occur on sufficiently short timescales that they are complete well before the chemical processes occur. In the case of a simple unimolecular



dissociation, this requirement leads to a steady-state distribution in the grain populations and to a unimolecular rate coefficient equal to $|\lambda_1|$, the eigenvalue of smallest magnitude (35). In some circumstances, however, the timescale separation between chemical and collisional relaxation modes is not so clear cut, and complications can arise, especially at low pressures and high temperatures.

A good example is provided by studies of the dissociation of the methyl radical in shock tubes (47). One of the product channels gives $\text{CH}_2 + \text{H}$, and H formation was observed by atomic resonance absorption spectroscopy. CH_3 was generated from the dissociation of acetone and of azomethane and, at low total pressures (~ 100 mbar) and in the temperature range 2400–3200 K, a delay of $\sim 100 \mu\text{s}$ was observed in the growth of the H signal. This “incubation time” was ascribed to formation of vibrationally cold CH_3 , because the unimolecular dissociation was close to the low-pressure limit, and to slow collisional energization, on a timescale comparable with the fragmentation of the CH_3 . Thus two timescales of comparable magnitude operate for this simple dissociation process, under these conditions. Eng et al. (47) obtained agreement with ME calculations, provided they modified the latter to include the lowest vibrational levels of CH_3 explicitly, i.e., they did not lump the states into energy grains for the lowest levels.

2.3. Isomerization

The model described in Section 2.2 applies to irreversible unimolecular reaction only, that is, to those reactions for which there is negligible contribution from the reverse reaction. Many reactions can be approximated in this way, and the isopropyl decomposition cited earlier is a typical example. However, the reverse reaction can make a significant contribution to the overall rate process requiring the ME model to be extended.

The simplest class of reversible reactions is isomerization, generally represented as,



MEs describing the energy transfer dynamics and reactive loss of both A and B are easily written, as they are simply Equation 23 applied separately to each isomer. Such a set of equations is incomplete because they do not conserve overall probability density, in contradiction to the conservation of mass implicit in a

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Figure 1 Fall-off curves for the decomposition of 2-propyl. The experimental rate coefficients were obtained using photoionization mass spectroscopy at the following temperatures: ×, 840 K; □, 830 K; +, 820 K; ◇, 810 K; ✱, 800 K; △, 790 K; ×, 780 K; □, 770 K; +, 760 K; ◇, 750 K. The lines are fits obtained using the ME/inverse laplace transform approach, with data for the reverse association reaction from Reference 39. The enthalpy of formation of 2-propyl and $\langle \Delta E \rangle_d$ were employed as variable fitting parameters.

reversible isomerization. This situation is rectified by adding reactive gain terms to Equation 23, so that the ME for isomer A becomes:

$$\begin{aligned} \frac{\partial}{\partial t} \rho^A(E, t) = & \omega \sum_{E'} P(E|E') \rho^A(E', t) - \omega \rho^A(E, t) \\ & - k^A(E) \rho^A(E, t) + k^B(E) \rho^B(E, t), \end{aligned} \quad 30.$$

where the superscripts indicate the isomer to which the associate term belongs. The reactive gain term reflects the observation that any density loss by isomer A must appear in isomer B and vice versa. The two sets of MEs are now coupled to give an extended ME that describes the dynamics of the overall isomerization reaction. This extended ME is still linear in the population density, which now encompasses the states of both isomers, and can be written in the same compact form as Equation 24. The general solution of the extended ME is the same eigenvalue/eigenvector expansion given in Equation 25. There are, however, important differences as there are now two eigenvalues that are well separated in magnitude from the others, one of which is exactly zero. The eigenvector corresponding to the zero eigenvalue is the equilibrium distribution for the isomers (i.e., Boltzmann distributions in each isomer, with the ratio of total density in each isomer equal to the equilibrium constant). This situation reflects the observation that the final state of the system must be the equilibrium distribution. The second eigenvalue is, as before, negative and dictates the evolution of the system over long timescales, the difference being that it now controls the approach to equilibrium rather than the decay of reactant.

Numerical solution of the isomerization ME proceeds as before, that is, a suitable basis set is used to express the collision operator, which is then diagonalized to extract the eigenvalues and eigenvectors. Again, the most common basis used to date is the δ -function basis, and it is interesting to examine the form of the collision matrix obtained, which is shown in Figure 2. The matrix is partitioned into blocks: blocks on the diagonal govern energy transfer and reactive loss for each isomer and are essentially the same as the EGME for a simple unimolecular dissociation reaction. The off-diagonal blocks deal with reactive gain and provide the coupling between the separate unimolecular blocks.

In Section 2.2 the unimolecular rate coefficient was identified as the eigenvalue of smallest magnitude by determining the total concentration and substituting into the phenomenological rate equation. The phenomenological rate equations for the isomerization reaction are:

$$\begin{aligned} \frac{d[A]}{dt} &= -k_f[A] + k_r[B] \\ \frac{d[B]}{dt} &= k_f[A] - k_r[B], \end{aligned} \quad 31.$$

where k_f and k_r are the rate coefficients for forward and reverse reaction respectively. To establish a similar relationship between k_f and k_r and the eigenvalues of the ME, it is useful to look first at the general solution to Equations 31, which

$$\begin{pmatrix} M_{11}^A & \omega P_{12}^A & \dots & \omega P_{1n}^A & 0 & \dots & 0 & 0 \\ \omega P_{21}^A & M_{22}^A & \dots & \omega P_{2n}^A & \vdots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \vdots & 0 & \dots & k_{l-1}^B & 0 \\ \omega P_{n1}^A & \omega P_{n2}^A & \dots & M_{nn}^A & 0 & \dots & 0 & k_l^B \\ 0 & \dots & 0 & 0 & M_{11}^B & \omega P_{12}^B & \dots & \omega P_{1l}^B \\ \vdots & \ddots & \vdots & \vdots & \omega P_{21}^B & M_{22}^B & \dots & \omega P_{2l}^B \\ 0 & \dots & k_{n-1}^A & 0 & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & k_n^A & \omega P_{l1}^B & \omega P_{l2}^B & \dots & M_{ll}^B \end{pmatrix}$$

Figure 2 The δ -function representation of the ME for an isomerization reaction: $M_{ii}^A = -[\omega(1 - P_{ii}^A) + k_i^A]$, n is the label of the highest energy grain for isomer A and the label of the highest grain B. P_{ij} is the probability of transfer on collision from grain j to grain i , and k_i^A is the rate coefficient for isomerization from grain i of isomer A to the isoenergetic grain of isomer B. ω is the collision frequency. This representation is the δ function equivalent of the continuous form given in Equation 30.

can be solved by an eigenvalue approach. If the vector $\mathbf{c}$ is defined as,

$$\mathbf{c} = \begin{pmatrix} [A] \\ [B] \end{pmatrix}. \quad 32.$$

Then Equation 31 can be written in a similar form to that of Equation 24:

$$\frac{d\mathbf{c}}{dt} = \mathbf{M}_c \mathbf{c}, \quad 33.$$

where $\mathbf{M}_c$ is given by 34.

$$\mathbf{M}_c = \begin{pmatrix} -k_f & k_r \\ k_f & -k_r \end{pmatrix}. \quad 34.$$

Diagonalization of $\mathbf{M}_c$ yields two eigenvalues, one of which is zero and the other is $-(k_f + k_r)$. The second eigenvalue dictates the rate at which the system reaches equilibrium. Comparing these eigenvalues with those obtained from the ME allows the following identification to be made,

$$-\lambda_2 = k_f + k_r. \quad 35.$$

Because $k_f/k_r = K_{eq}$, the equilibrium constant, k_f and k_r are given by:

$$\begin{aligned}
 k_f &= \frac{-\lambda_2}{1 + 1/K_{eq}} \\
 k_r &= \frac{-\lambda_2}{1 + K_{eq}}.
 \end{aligned}
 \tag{36}$$

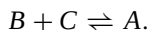
As before, Equations 36 are only valid when λ_2 is well separated from the higher eigenvalues obtained from the ME. Even when this restriction is satisfied, some workers (48) have questioned the validity of these equations on the basis of the conventional definition of the canonical rate coefficient as the canonical average of the microcanonical rate coefficients, i.e.,

$$k'_f = \frac{\sum_{i \in A} k_i^A \rho_i^{A(t)}}{\sum_{i \in A} \rho_i^{A(t)}}.
 \tag{37}$$

In Equation 37, the time dependence of the density does not factor out simply, and as a consequence k'_f is time dependent, unlike k_f . The problem is that the definition of the rate coefficient in Equation 37 does not take account of back reaction. Green et al. (49) investigated this issue and concluded that, provided back reaction is properly accounted for, then k_f and k_r as defined in Equations 36 do describe the approach to equilibrium correctly on the important experimental timescale. The interpretation of k_f and k_r , however, is different from that implied by Equation 37— k_f is the rate coefficient for the irreversible process that occurs when an absorbing boundary is inserted into the state space of isomer B below the reaction threshold. Insertion of an absorbing boundary in this way allows back reaction to be accounted for, whereas Equation 37 implies that the absorbing boundary is placed between reactants and therefore does not allow for back reaction.

2.4. Association

Another important class of reactions that frequently occurs in combustion and atmospheric systems is association, generally represented in reversible form as,



The inherent difficulty with these reactions is that they are nonlinear and so it is difficult to write down and solve a ME.

Usually, the quantity of principal interest is the association rate coefficient and its temperature and pressure dependence. Early attempts at applying ME techniques to these reactions involved writing down and solving the ME for the dissociation reaction to give the dissociation rate coefficient and then applying the equilibrium constant to obtain the association rate coefficient:

$$K_{eq} = \frac{k_a}{k_u} = \frac{[A]_{eq}}{[B]_{eq}[C]_{eq}},
 \tag{38}$$

where, K_{eq} is the equilibrium constant, k_a is the effective forward, association rate coefficient and k_u is the complementary unimolecular dissociation rate coefficient.

This approach can present numerical difficulties, especially at low temperatures, and has motivated a number of workers to pursue alternative approaches. One such approach is to modify the standard unimolecular ME (Equation 24) by adding a source term (38):

$$\frac{\partial}{\partial t} \rho(E, t) = \hat{M} \rho(E, t) + g(E, t). \quad 39.$$

The source term, $g(E, t)$, represents the rate of reactive gain owing to association. To deduce its functional form we note that integrating with respect to E gives the overall association rate, i.e.,

$$\int g(E, t) dE = k_{a,\infty} [B](t) [C](t), \quad 40.$$

where $k_{a,\infty}$ is the limiting high pressure rate coefficient for association. If it is assumed that the energy and time dependence can be separated, then

$$g(E, t) = R(t) \phi(E) = k_{a,\infty} [B](t) [C](t) \phi(E), \quad 41.$$

where the function $\phi(E)$ distributes the input rate among the energies of A . By the application of detailed balance it can be shown that

$$\phi(E) = \frac{k(E) f(E)}{\int k(E) f(E) dE}. \quad 42.$$

Equation 41 indicates the origin of the difficulty in treating these systems using the ME approach, in that $R(t)$ is, in general, nonlinear. One approximate approach to solving Equation 41 is to introduce an absorbing boundary into the grains of A in a similar way to that described earlier for isomerization systems. The absorbing boundary is usually placed about $10 k_B T$ below the reaction threshold and has the effect of making the system irreversible; after a short period the solution for the population density converges to a steady state, which has the solution

$$\rho_s(E, t) = -R(t) \hat{M}^{-1} \phi(E). \quad 43.$$

This is a steady-state solution in the sense that the relative population of each energy state is constant while the absolute value decays as the reactants are consumed. The steady-state density can be used to calculate the rate of dissociation to regenerate the reactants, which can then be subtracted from the total input rate to give the effective forward rate of association, from which the rate coefficient can be obtained as

$$k_a = k_{a,\infty} \left(1 + \int k(E) \hat{M}^{-1} \phi(E) dE \right). \quad 44.$$

Alternatively, the association rate coefficient can be obtained from the rate into the absorbing barrier. These approaches are useful if the association rate coefficient is all that is required. It should be noted that the solution is time-dependent and is generated for a specific instantaneous value of the product $[A][B]$ (usually taken as unity). The method has been employed in fits to rate data for the irreversible

association reaction, $\text{CH}_3 + \text{CH}_3$, using a wide range of data sources, covering an extensive range of temperatures (38). Inverse laplace transformation of the limiting high pressure association rate coefficient was again employed. The data were fitted to the form $AT^n \exp(B/T)$, with A determined from the well-defined rate coefficient at 300 K, coupled with the fitted values of n and B . This non-Arrhenius form was found to be essential to reproduce the experimental data, but no unique solution to the two variable parameters could be found, reflecting either inconsistencies or inadequacies in the experimental database. One problem is that, at high temperatures, the experimental data are far from the high pressure limit over the whole experimental pressure range, so that unique extrapolations are not feasible.

More general solutions to Equation 39 have been investigated by Smith et al. (50) and, more recently, by Knyazev & Tsang (51), both giving a more detailed solution in terms of a convolution involving the source term

$$\rho(E, t) = \exp(\hat{M}t)\rho(E, 0) + \int_0^t R(s) \exp(\hat{M}(t-s))\varphi(E) ds. \quad 45.$$

The difficulty with this solution is that the time dependence of $R(t)$ must be known a priori, however this in turn is intimately related to that of $\rho(E, t)$, as must be the case if the system is reversible and is approaching equilibrium, and so approximation is required to obtain a solution from Equation 45.

One situation in which a complete solution can be obtained is when one of the reactants is in substantial excess, such that reaction makes little impact on its concentration. Pseudo-first-order conditions pertain and a ME is readily formulated. The solution of this ME proceeds in exactly the same way as before, in that a matrix representation of $\hat{M}$ in some suitable basis is constructed and diagonalized to obtain the eigenvalues. Thus, if the excess reactant is C, a typical matrix using a δ -function basis takes the form

$$\begin{pmatrix} M_{11}^A & \cdots & M_{1n}^A & k_1 f_1 K_{eq}[C] \\ \vdots & \ddots & \vdots & \vdots \\ M_{n1}^A & \cdots & M_{nn}^A & k_n f_n K_{eq}[C] \\ k_1 & \cdots & k_n & -k_{a,\infty}[C] \end{pmatrix}. \quad 46.$$

This matrix acts on a population vector whose first n elements are the grain populations of A and whose $(n+1)$ grain is the total population of the reactant that is not in excess, in this case B. Implicit in this approach is that only the time dependence of the overall concentration of B is important. Effectively the assumption is made that B conforms to a Boltzmann distribution, although it is possible to extend this approach to include individual states of B.

Because the system is now conservative, one eigenvalue is zero and the rest are negative; it is usually found that the nonzero eigenvalue of smallest magnitude differs from the others significantly and thus governs the relaxation to equilibrium.

Rate coefficients for forward and reverse reactions can be obtained by applying an analysis similar to that used for the isomerization case, giving,

$$\begin{aligned} k_a &= \frac{-\lambda_2}{[C] + 1/K_{eq}} \\ k_u &= \frac{-\lambda_2}{1 + K_{eq}[C]}. \end{aligned} \quad 47.$$

At this point, and by way of introduction to Section 3, it is of interest to examine a specific association reaction under experimental conditions, by reference to a laser flash photolysis study of $H + C_2H_4$ under both low and high temperature conditions (52). At low temperatures, the reaction is irreversible and the absorbing boundary approach can be used. At high temperatures, feedback from the dissociation of C_2H_5 occurs, so that both H and C_2H_5 are included in the vector of chemical species, $\mathbf{c}$ (Equation 32). The reactions and rate equations, under experimental conditions, are:



$$\begin{aligned} \frac{da}{dt} &= -(k'_1 + k_2)a + k_{-1}b \\ \frac{db}{dt} &= -k_{-1}b + k'_1a, \end{aligned} \quad 48.$$

where $a = [H]$, $b = [C_2H_5]$, k'_1 is the pseudo-first-order rate coefficient for the association of H and C_2H_4 , with $[H] \ll [C_2H_4]$, k_{-1} is the rate coefficient for dissociation of C_2H_5 and k_2 for loss of H by other reactions. Under the experimental conditions employed, C_2H_5 only underwent Reaction R-1, although other first-order loss processes can be readily incorporated into the analysis. The rate equations can be expressed in matrix form as:

$$\frac{d\mathbf{c}}{dt} = \mathbf{M}_c \mathbf{c}, \quad 33.$$

where

$$\mathbf{M}_c = \begin{pmatrix} -(k'_1 + k_2) & k_{-1} \\ k'_1 & -k_{-1} \end{pmatrix}. \quad 49.$$

The solution of Equation 33 follows closely that described for isomerization above, and the eigenvalues of $\mathbf{M}_c$, λ_1 , and λ_2 are given by

$$\lambda_{1,2} = -\frac{k_T}{2} \left\{ 1 \pm \sqrt{1 - (4P/k_T^2)} \right\}, \quad 50.$$

where $k_T = k'_1 + k_{-1} + k_2$ and $P = k_{-1}k_2$. As with the isomerization example, these eigenvalues can be equated with the two eigenvalues of smallest magnitude obtained from the matrix representation of the ME operator $\hat{M}$ of this system allowing values of the rate coefficients to be extracted. Note that the presence of the additional reaction term in k_2 complicates the analysis, so that the chemical system now evolves as a biexponential. Provided these modes are well separated in time from the collisional energy relaxation modes, the matrix $\mathbf{M}_c$ again describes the chemical evolution of the system. $\mathbf{M}_c$ corresponds to the Jacobian of the set of coupled rate equations of the chemical system (53). The rates of the coupled set of chemical reactions (Equation 50) can be expressed in the form:

$$\frac{d\mathbf{c}}{dt} = f(\mathbf{c}, \mathbf{k}), \quad 51.$$

where $\mathbf{k}$ is a vector of rate coefficients. The elements of $\mathbf{M}_c$ are given by:

$$M_{i,j} = \frac{\partial f_i}{\partial c_j}. \quad 52.$$

The diagonal elements of $\mathbf{M}_c$ correspond to the negated first-order or pseudo first-order rate coefficients for each species, i.e., to the negated $1/e$ or chemical lifetimes of the species.

From Equation 50, eigenvalues corresponding to the chemical relaxation modes of the system can be expressed as functions of the phenomenological rate coefficients. It is instructive to examine limiting cases for the $\text{H} + \text{C}_2\text{H}_4$ system. At low temperatures, and with the association reaction dominating the loss of H, $k_1 \gg k_{-1}$, k_2 , so that the square root can be expressed as a Taylor expansion, truncated after the second term, giving $\lambda_1 \approx -\frac{k_{-1}k_2}{k_1} = k_2/K_1$ and $\lambda_2 \approx -k_1$. Under these conditions, and following pulsed photochemical generation of H, the system evolves by production of ethyl from H with eigenvalue $\lambda_2 = -(k_1 + k_2) \approx -k_1$. At long times the relaxation is given by the rate coefficient for loss from the system, k_2 , multiplied by the fraction of the system in the form of the species being lost, i.e., H. Provided the temperature is sufficiently low that the reaction is essentially irreversible, $1/K_1 \approx 0$, and $\lambda_1 \approx 0$. Similarly, if k_2 is zero, then the system is conservative and $\lambda_1 = 0$. $\lambda_2 = -(k_1 + k_{-1})$, which is the negated reciprocal relaxation time for approach to equilibrium. The rate coefficients can be determined, once again, from Equation 47.

The eigenvalues are independent of the initial conditions so that we may also use $\mathbf{M}_c$ to consider the dissociation reaction at high T, with $[\text{H}]_{t=0} = 0$. If $[\text{C}_2\text{H}_4]$ can be considered small, so that $k_1 \approx 0$, then the higher chemical eigenvalue, λ_2 , switches and becomes equal to $-k_{-1}$, the negated reciprocal chemical relaxation time for C_2H_5 .

These ideas can be readily extended to more complex chemical systems, with several adduct species involved, corresponding to several wells in the potential energy surface, to several components in the chemical vector $\mathbf{c}$, and to an equal

number of chemical eigenvalues. Section 3 examines examples of such systems. Three general regions of simplification can be identified:

1. Provided the chemical eigenvalues are well separated in magnitude, they can be expressed as simple sums or products/ratios of rate coefficients and the chemical relaxation processes related to identifiable single step or coupled chemical processes.
2. If some of the eigenvalues are comparable in magnitude, however, a mixing occurs and this simple resolution is not possible. In the case of $\text{H} + \text{C}_2\text{H}_4$, Equation 50 cannot now be simplified in the ways that were discussed above. Provided the mixing is not too great, first-order perturbation theory can be employed to understand the nature of the mixing. For $\text{H} + \text{C}_2\text{H}_4$, as the temperature is increased and the equilibration regime is approached from below, the eigenvalues become:

$$\begin{aligned} |\lambda_2| &= k_{\text{H}} + \frac{k_1 k_{-1}}{k_{\text{H}} - k_{\text{Et}}}, \\ |\lambda_1| &= k_{\text{Et}} - \frac{k'_1 k_2}{k_{\text{H}} - k_{\text{Et}}}, \end{aligned} \tag{53}$$

where $k_{\text{H}} = k'_1 + k_2$ and $k_{\text{Et}} = k_{-1}$, the reciprocal lifetimes of the two time-dependent species, H and C_2H_5 . Thus the extent of the perturbation of the moduli of the eigenvalues from the appropriate reciprocal lifetimes depends on the strength of the off-diagonal coupling terms, k_1 and k_{-1} , and on the difference in the reciprocal lifetimes. If this difference is substantial, then the perturbation is small and only becomes significant as the terms become comparable.

3. If $\mathbf{c}$ is ordered such that the shortest-lived species is placed first, and if there is negligible feedback from longer-lived to shorter-lived species, then $\mathbf{M}_{\mathbf{c}}$ becomes lower triangular, with all of the elements above the diagonal equal to zero. In the $\text{H} + \text{C}_2\text{H}_4$ case, k_{-1} approaches zero. Under these circumstances, the eigenvalues of the matrix become equal to the diagonal elements, i.e., to minus the reciprocal lifetimes.

Although these arguments are self-evident in the simple case discussed above, they are very useful tools for more complex systems. Klippenstein & Miller (54) examined the relationship between the eigenvalues of the ME and the rate coefficients of the chemical mechanism in some detail, demonstrating the power of the approach by reference to the $\text{C}_2\text{H}_3 + \text{C}_2\text{H}_2$ reaction. They showed that the rate coefficients can be expressed explicitly as summations over components of the products of the eigenvalues and eigenvectors. They also employed expressions from a different and more general approach taken by Bartis & Widom (55).

3. APPLICATIONS OF THE MASTER EQUATION

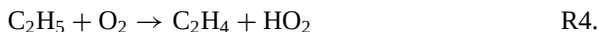
3.1. $\text{C}_2\text{H}_5 + \text{O}_2$

At low temperatures, the reaction between ethyl and O_2 involves association to form the peroxy radical (Reaction R3). As the temperature increases to ~ 600 K, the reverse decomposition of the peroxy radical becomes increasingly important and the equilibrium swings increasingly to the left (Reaction R3,-3).



The ceiling temperature, T_c , is defined as that temperature at which $[\text{C}_2\text{H}_5] = [\text{C}_2\text{H}_5\text{O}_2]$ at equilibrium for an oxygen pressure of 0.1 atm. For ethyl, $T_c = 900$ K (56).

An additional reaction channel occurs, which leads to formation of $\text{C}_2\text{H}_4 + \text{HO}_2$:



Reaction R4 occurs even at room temperature and at low pressures, but it is suppressed in favor of Reaction R3 with increasing pressure. At high temperatures, Reaction R4 dominates the overall reaction. The reaction has been studied extensively by Walker and coworkers (8), using classical end-product analysis, by Gutman and coworkers (14, 57) using laser flash photolysis, coupled with photo-ionization mass spectrometry, by Kaiser (10–13) using end-product analysis, and by Clifford et al. (9) using laser flash photolysis and infrared (IR) laser-absorption spectroscopy. The investigations together provide an extensive body of high quality experimental data that has been central to the elucidation of the mechanism. A crucial step was the interpretation of the Gutman experiments, which provided a coherent description based on a route from energized $\text{C}_2\text{H}_5\text{O}_2$ to $\text{C}_2\text{H}_4 + \text{HO}_2$, contrasting with previous suggestions that this reaction involves simple H abstraction.

The potential energy surface has been studied in great detail by Schaefer et al. (15–17). The latest analysis uses high-level coupled-cluster methods [CCSD and CCSD(T)], and shows that the lowest energy route for Reaction R4 involves concerted elimination of HO_2 from the ethyl peroxy radical, with a transition state at or slightly below the energy of reactants. They found that a mechanism involving the initial formation of a hydroperoxy radical by β H atom transfer, followed by C–O bond rupture, has a significantly higher energy barrier, as does the dissociation of the hydroperoxy radical to form oxirane + OH.



Stark (18) addressed the issue of the reaction between HO_2 and C_2H_4 , which produces oxirane + OH rather than $\text{C}_2\text{H}_5 + \text{O}_2$ (58, 59) and suggested that this reaction proceeds via the hydroperoxy radical, which is accessed by a curve-crossing mechanism, involving the first excited electronic state.

Miller et al. (19, 20) examined the $\text{C}_2\text{H}_5 + \text{O}_2$ reaction in detail using a ME analysis, restricted, in the main, to Reactions R3, -3, 4 and under pseudo first order conditions ($[\text{O}_2] \gg [\text{C}_2\text{H}_5]$). They used density functional theory (DFT-B3LYP) to calculate the characteristics of the overall potential energy surface, with G2 theory to improve the energies of the turning points. They showed that the reaction can be characterized by three different regimes:

1. At low temperatures ($T < 575 \text{ K}$), the system can be described by Reactions R3 and R4. The overall rate coefficient is pressure-dependent, and the efficiency of formation of $\text{C}_2\text{H}_4 + \text{HO}_2$, $\alpha = k_4/(k_{-3} + k_4)$ decreases with increasing pressure in the expected way. At any given pressure, α increases with temperature. Note that in this temperature regime, the peroxy radical is stable on the timescale of the $\text{C}_2\text{H}_5 + \text{O}_2$ reaction. The $\text{C}_2\text{H}_4 + \text{HO}_2$ reaction route derives entirely from reaction from energized $\text{C}_2\text{H}_5\text{O}_2$, formed from the reactants $\text{C}_2\text{H}_5 + \text{O}_2$. The ME can be satisfactorily formulated using an absorbing boundary to describe the formation of $\text{C}_2\text{H}_5\text{O}_2$, so that there is only a single eigenvalue, corresponding to loss of C_2H_5 .
2. At intermediate temperatures, ($575 \leq T/\text{K} \leq 750$), the system becomes more complex, because both the reverse Reaction (R-3) and dissociation of thermalized $\text{C}_2\text{H}_5\text{O}_2$ to form $\text{C}_2\text{H}_4 + \text{HO}_2$ become significant. Under these conditions, it is necessary to include all of the $\text{C}_2\text{H}_5\text{O}_2$ grains in the ME and to include $\text{C}_2\text{H}_5\text{O}_2$ in the phenomenological description of the chemistry. In temperature regime 1, this could be achieved using an uncoupled description of the $\text{C}_2\text{H}_5\text{O}_2$ chemistry. In regime 2, the peroxy radical has to be explicitly recognized, because the chemical lifetimes of C_2H_5 and $\text{C}_2\text{H}_5\text{O}_2$ are now becoming comparable. From the ME viewpoint, the two eigenvalues of lowest magnitude are now related to overall chemical processes.

In this regime, the two eigenvectors become mixed and a full description requires an analysis similar to that given above for $\text{H} + \text{C}_2\text{H}_4$. Clifford et al. (9) have studied this regime experimentally, through observations of HO_2 by IR laser absorption. They found two timescales for HO_2 production. One, which they termed “prompt,” derives from the formation of HO_2 from $\text{C}_2\text{H}_5\text{O}_2^*$, formed directly from $\text{C}_2\text{H}_5 + \text{O}_2$; the other relates to formation from dissociation of thermalized $\text{C}_2\text{H}_5\text{O}_2$. This phenomenological description becomes less satisfactory as the temperature increases—the eigenvalues are mixed and the description more complex.

3. At higher temperatures, $T/\text{K} \geq 750$, the system can once again be described by a single eigenvalue. In this regime, stabilization of $\text{C}_2\text{H}_5\text{O}_2^*$ is ineffective at all realizable pressures, and the sole products are $\text{C}_2\text{H}_4 + \text{HO}_2$. The lowest eigenvalue now describes Reaction R4 and a “zero pressure” ME model now provides an adequate description of the system—stabilization is significantly slower than reaction via Reaction R4.

Miller & Klippenstein (20) defined a “stabilization” limit in this context. In this limit, stabilization of $\text{C}_2\text{H}_5\text{O}_2^*$ is ineffective, “no matter how high one may

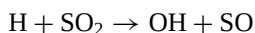
make the pressure.” This limit marks the transition between the intermediate- and high-temperature regimes.

Some further comments on this reaction system are appropriate.

Reaction R5 is comparatively ineffective in the C_2 system, because it involves a transition state with a five-membered ring, which, according to both the G2 calculations of Miller et al. (19), and the higher level calculations of Rienstra-Karicofe et al. (17), is too high in energy to play a significant role. Miller & Klippenstein calculated the efficiency of the OH channel and obtained a value of 0.01 at 800 K, in good agreement with the value obtained experimentally by Baldwin et al. (60). For larger alkyl radicals, however, this reaction becomes more significant and provides an important propagation route in alkane oxidation, and in autoignition chemistry, via OH formation. For neopentane (2,2 dimethyl propane), this is the only high-temperature channel, because the absence of β hydrogen precludes the $C_5H_{10} + HO_2$ channel. An extensive set of experimental measurements by Baldwin, Walker, and coworkers (56) demonstrate clearly and quantitatively the importance of these channels in the autoignition chemistry of higher alkanes.

Miller & Klippenstein also commented on the importance of the stabilization limit on the temporal evolution of the $C_2H_5 + O_2$ system. Alkane oxidation shows a negative temperature regime where the rate of reaction decreases as the temperature is increased (56). The explanation of this phenomenon has centred around the importance of the reversal of the position of the equilibrium in Reaction R3, coupled with the consequent onset of Reaction R4 at higher T . Miller & Klippenstein ascribed this effect entirely to the onset of the stabilization limit. Although their argument is sound for the C_2 system, its extension to higher hydrocarbons requires further analysis.

3.2. $H + SO_2$



has been identified as a significant sink for SO_2 in combustion systems. Examples include the low-pressure flame experiments of Hughes et al. (61), where $CH_4/O_2/Ar$ mixtures were doped with NH_3 and SO_2 . Hughes et al. also found that the reaction couples the SO_x and NO_x chemistry, which is an important and poorly understood issue in pollutant control in combustion.

The reaction is endothermic and proceeds via the isomers HSO_2 and $HOSO$



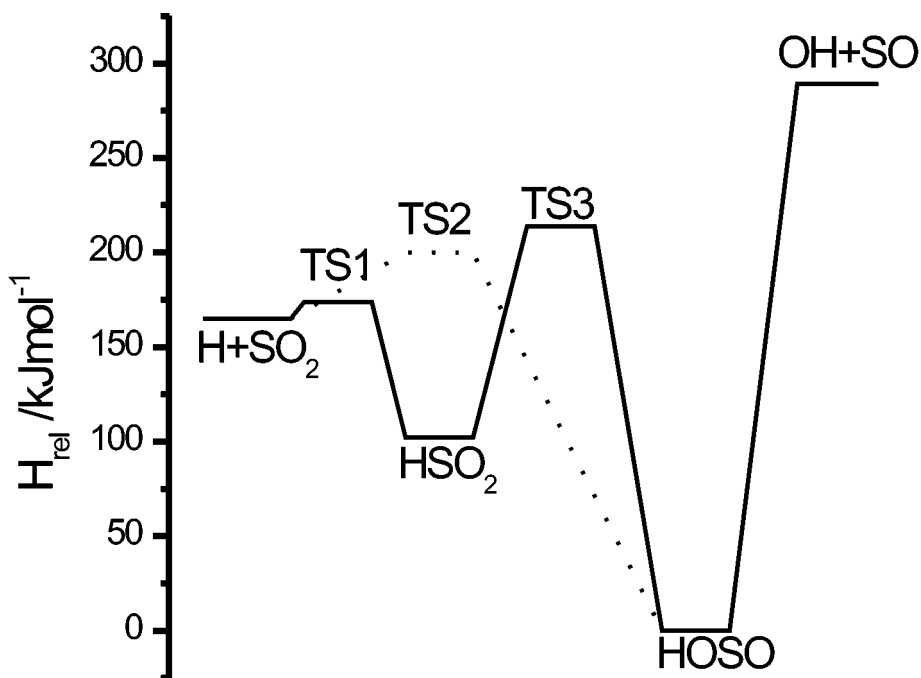
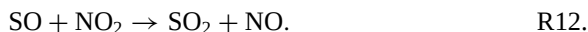


Figure 3 Schematic potential energy surface for $\text{H} + \text{SO}_2$, based on the calculations of Goumri et al. (62).

The potential energy surface for the reaction system, calculated by Goumri et al. (62), is shown in Figure 3. A similar surface has been calculated by Frank et al. (63).

The reaction has not been well characterized experimentally. Blitz et al. (64) made the most detailed investigation of the reverse reaction, using laser flash photolysis coupled with laser-induced fluorescence detection of OH, and covering the temperature range 295–703 K, with $[\text{OH}] \ll [\text{SO}]$. SO was formed by photolysis of SO_2 or SOCl_2 and $[\text{SO}]$ was determined by adding NO_2 in separate experiments and monitoring by laser induced fluorescence (LIF) the NO formed in the reaction



They obtained a strong negative-temperature dependence for the rate coefficient, which they ascribed to the reverse of Reaction R9 and expressed as $k_{-9} = (8.3 \pm 0.4) \times 10^{-11} (\text{T}/295 \text{ K})^{-1.35 \pm 0.11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$.

The relationship between the rate constants for the $\text{OH} + \text{SO}$ and $\text{H} + \text{SO}_2$ reactions is complex, because of the involvement of the intermediates HOSO and HSO_2 . Indeed these species are included explicitly in the combustions mechanisms developed by Glarborg and coworkers (65). Hughes et al. (66) investigated the system using a ME analysis. They used the concatenated single population

vector approach, discussed in Section 2.3; the vector contained the normalized grain populations of HSO_2 and HOSO and the population of H ; the calculations were conducted with a fixed, high concentration of SO_2 (10^{15} cm^{-3}) to ensure linearity of the ME. The formation of $\text{OH} + \text{SO}$ was treated as irreversible, so that the system was not conservative and all eigenvalues were nonzero. They used the transition-state properties from the ab initio surface of Goumri et al. (62) to calculate the microcanonical rate coefficients for the constrained transitions states (TS1-3), (Figure 3), and inverse laplace transformation on the measured $k_{12}(T)$ for the dissociation of HOSO to form $\text{OH} + \text{SO}$. They summed the grain populations of HSO_2 and HOSO to obtain fractional concentrations for these species, whereas $[\text{H}]$ was obtained from its specific vector element; $[\text{OH}]$ was obtained from the normalization condition. Calculation of the temporal evolution of the concentrations was made over the temperature and pressure ranges 300–2000 K and 10^{-3} – 10^6 atm, respectively. Figure 4 shows a representative example. An interesting feature is the formation of $\text{OH} + \text{SO}$ from $\text{H} + \text{SO}_2$, despite the absence of a transition state for this direct transition. This channel is described phenomenologically by Reaction R9 and corresponds microscopically to the formation of products from

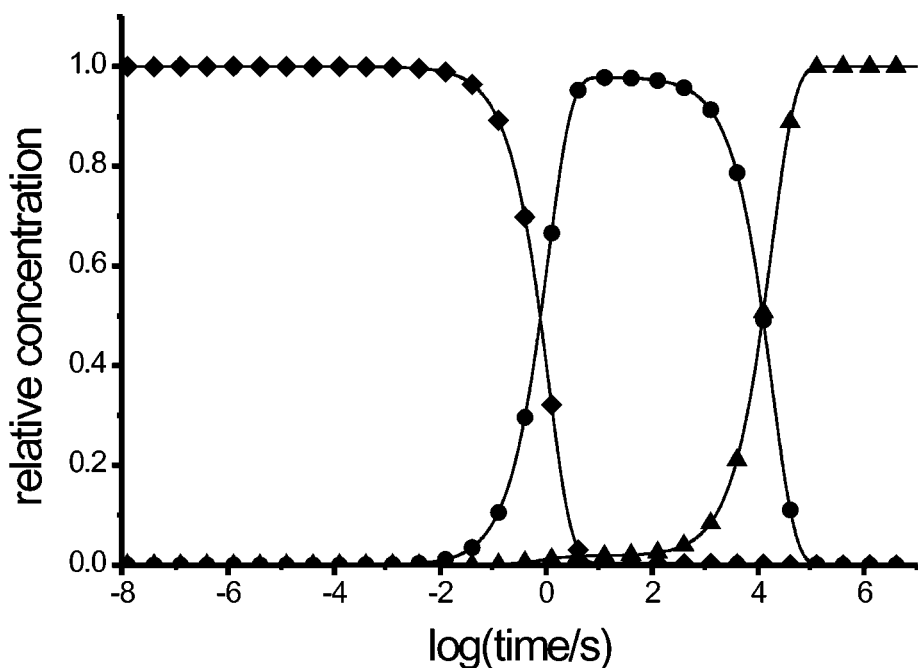


Figure 4 Evolution of the normalized concentrations of H (◆), HOSO (●) and OH (▲) at 800 K and 1 Torr. The symbols refer to the results of the ME and the lines to the fits to the phenomenological mechanism, based on the concentrations derived from the ME.

the energized grains of HOSO, without collisional stabilization and subsequent collision induced dissociation.

Hughes et al. (66) fitted the concentrations calculated via the ME to analytic expressions obtained from the phenomenological chemical model (Reactions R7–R11). The full reaction set has three independent species (H, HSO₂, and HOSO), giving a tri-exponential, but this can be approximated, with negligible loss of accuracy, to a low temperature mechanism (neglecting formation of OH and obtaining HOSO by difference, i.e., omitting Reactions R 9 and R11) and a high-temperature mechanism (neglecting HSO₂ because the equilibrium for Reaction R7 is strongly to the left, i.e., omitting Reactions R7 and R10), resulting in biexponentials. The fitting process allowed determination of the rate coefficients for Reactions R7–R11, which were then fitted to modified Troe expressions (67). Hughes et al. also examined the evolution of the eigenvalues, comparing the dominant rate coefficients/rate coefficient combinations from the bi-exponential expressions with the moduli of the three lowest eigenvalues from the ME. This analysis complements the information provided by the temporal analysis of the concentration profiles. The following comments relate specifically to calculations performed at a pressure of 1 atm.

1. At all temperatures, $|\lambda_3| = k_7[\text{SO}_2] + k_{-7}$, the reciprocal relaxation time for Reaction R7,-7. For $T < 300$ K, $k_7[\text{SO}_2]$ dominates, but this term decreases as T increases and the reaction moves further into the fall-off region. In the temperature range 300–400 K, the pseudo-first-order rate coefficients for forward and reverse reactions are comparable and the reaction approaches equilibrium. At higher temperatures, k_{-7} dominates and $[\text{HSO}_2]$ becomes negligible; $|\lambda_3|$ remains equal to k_{-7} throughout this regime, but is now kinetically insignificant for the $\text{H} + \text{SO}_2$ reaction.
2. $|\lambda_2|$ is related, throughout the temperature range, to Reaction R8 (Figure 5). For $T < 400$ K, it is equal to $k_8[\text{SO}_2]$ multiplied by $k_{-7}/k_7[\text{SO}_2]$, the fraction of $([\text{H}] + [\text{HSO}_2])$ present as $[\text{H}]$. At higher T , this fraction becomes unity as HSO₂ decomposition to regenerate H becomes rapid. At around 1000 K, HOSO begins to dissociate at a significant rate and Reaction R8 equilibrates, so that k_{-8} begins to contribute significantly to $|\lambda_2|$, and eventually dominates as HOSO decomposition becomes faster and $[\text{HOSO}]$ becomes small. For temperatures greater than about 1200 K at 1 atm, Reaction R8 is essentially fully reversed and now plays little role in the overall reaction.
3. $|\lambda_1|$ relates to Reaction R9 and is essentially equal to $k_9[\text{SO}_2]$; below 1000 K, a factor $k_{-8}/k_8[\text{SO}_2]$ is necessary to correct for the fraction of the system present as HOSO.

The characteristic feature of this reaction is the dominance of the rapid dissociation reactions of both HOSO and HSO₂ at high temperatures, which severely limit the rate of the reaction. The reaction rates are too slow to account for the rate of loss of SO₂ in combustion systems, which may be caused by inadequacies in the potential energy surface. Alternatively Tsuchiya et al. (68) have recently

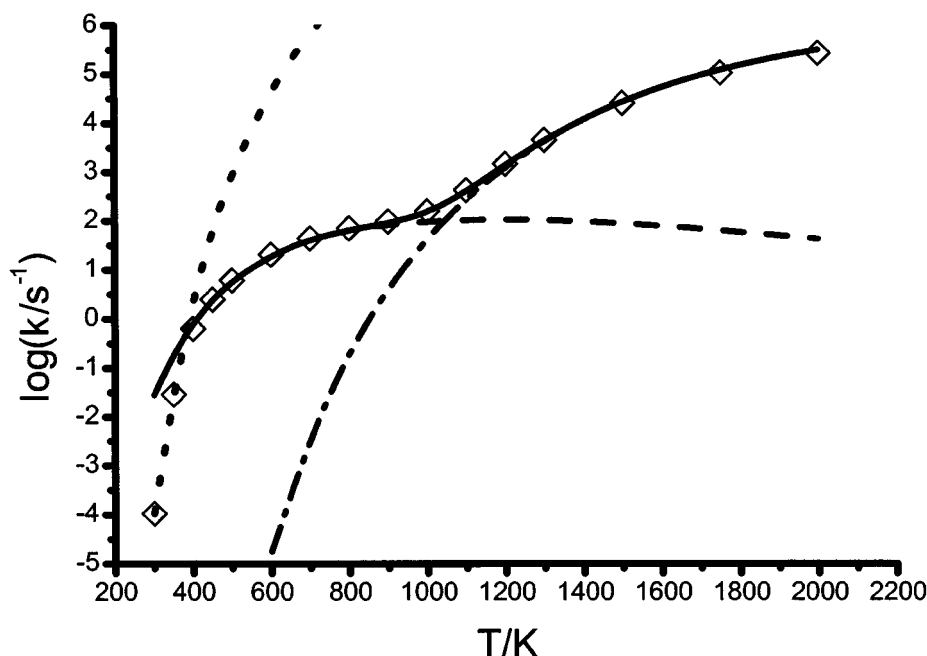


Figure 5 Dependence of $|\lambda_2|$ on temperature and the evolution of its relationship to the phenomenological rate constants for the $\text{H} + \text{SO}_2$ system. $\diamond$: eigenvalues. $\cdots$: $k_8 k_{-7}/k_7$; $---$: $k_8[\text{SO}_2]$; $- \cdot - \cdot -$: k_{-8} ; $—$, $k_8[\text{SO}_2] + k_{-8}$.

suggested, on the basis of ab initio calculations, that the reaction

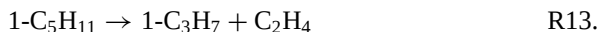


is rapid and this route may provide the step from HOSO to SO in combustion systems.

Blitz et al. (M.A. Blitz, K.J. Hughes, M.J. Pilling, S.H. Robertson, unpublished manuscript) have recently started to characterize the reaction system experimentally. Thus far, they have only investigated Reaction R7, using laser flash photolysis, coupled with vacuum ultraviolet (vuv) LIF to detect H. They found it necessary to tune the potential energy surface by increasing the activation energy for step R7 from 8.9 to 13.6 kJ mol⁻¹ to agree with the experimental data.

3.3. Alkyl Radical Decomposition and Isomerization

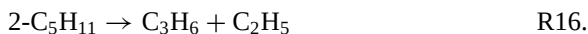
Alkyl radicals dissociate by either C–C or C–H bond fission to form alkenes and either smaller alkyl radicals or H atoms, e.g.,



Isomerization can also occur, by H atom transfer, for example the 1,4 H transfer in Reaction R15.



coupled with



The activation energies are 128, 89, 132 kJ mol⁻¹ for Reactions R13, 15, and 16, respectively.

Alkyl radical dissociation and isomerization reactions play a significant role in autoignition chemistry since, at the temperatures relevant to spark ignition engines, these reactions can compete with the reaction with O₂, discussed above for C₂H₅. Because the autoignition characteristics depend on the nature of the radical reacting with O₂, this competition is significant and is an important component of combustion models.

Tsang et al. investigated the coupled Reactions R13, R15, and R-15,16 (70) using a ME model and demonstrated that the activation energies are sufficiently small that the collisional and reactive eigenvalues overlap. Jitariu et al. (L.C. Jitariu, S.H. Robertson, I.H. Hillier, M.J. Pilling, unpublished manuscript.) considered a reaction system in which the initial reactant is 1-C₅H₁₁, which is rapidly heated from room temperature, e.g., in a shock tube. The mechanism can be reduced to Reactions R13, R15, and R-15, 16, because the high activation energy makes R14 uncompetitive and the corresponding ME is similar to that discussed for isomerization in Section 2.2, with the addition of dissociation terms along the main diagonal. The rate of Reaction R13, for example, can be calculated from sums of the sort $\sum k(E)\rho^A(E, t)$, where A is 1-C₅H₁₁.

The chemical reaction matrix, **M_c** is given by:

$$\mathbf{M}_c = \begin{pmatrix} -(k_{13} + k_{15}) & k_{15} \\ k_{-15} & -(k_{-15} + k_{16}) \end{pmatrix} \quad 55.$$

From this matrix it is evident that the temporal evolution of the chemical system is described by a bi-exponential with the reciprocal time constants given by the moduli of the two eigenvalues of smallest magnitude in the ME. This situation applies at temperatures below 1000 K, but at higher temperatures, λ₂ becomes comparable in magnitude with λ₃ (see Figure 6), so that this simple description is no longer appropriate. This situation presents problems in combustion modelling, because any attempt to use the two variable chemical model described by Equation 55 requires time-dependent rate coefficients. Such a situation is untenable, especially because the time dependence is not easily represented and depends on a range of reaction system variables. At higher temperatures (above ~1200–1400 K, depending on the pressure), the reactions occur much more rapidly than any competing alkyl radical reactions, so that only the product yields for Reactions R13 and R16 are required and these are easily calculated and represented in a form

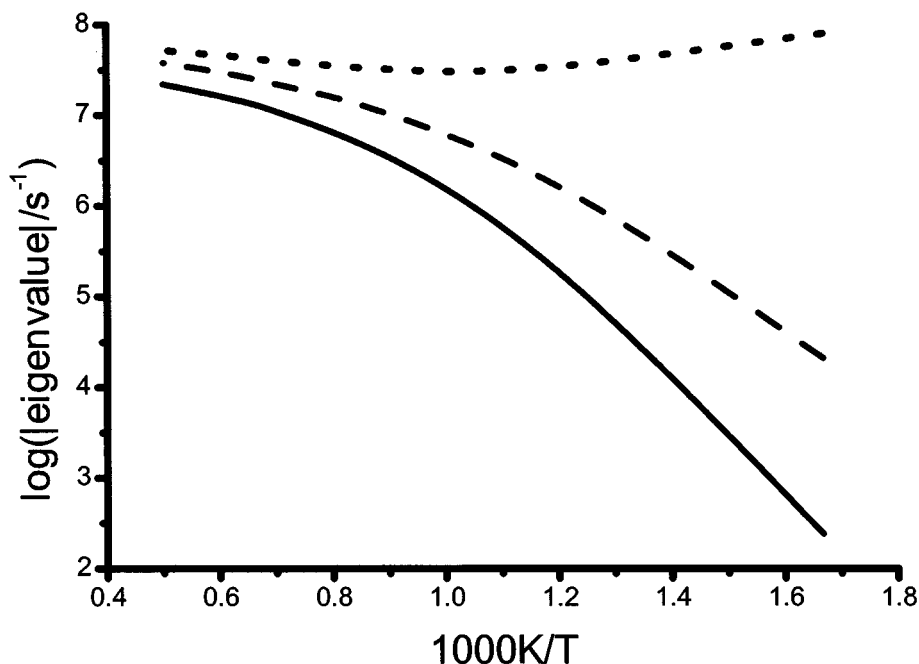


Figure 6 Dependence of $|\lambda_1|$, $|\lambda_2|$ and $|\lambda_3|$ on temperature for the 1,2 pentyl radical system. Pressure = 1000 torr.

appropriate for combustion modelling. The intermediate pressure range, though, presents as yet unresolved difficulties.

4. CONCLUDING REMARKS

The main aim of this review has been to outline the basis of the ME and to illustrate its use in the interpretation and representation of a range of reactions of importance in combustion. Recent years have seen considerable broadening in the range of applications of the ME, as demonstrated in Section 3. ME analysis is increasingly used in conjunction with *ab initio* calculations and in the analysis of experimental data, although such applications are still far from routine. The ideal approach is to use data from both experiment and quantum calculations to constrain the ME. Some quite complex reaction systems can now be tackled and there is much better understanding of the linkage between the eigenvalues determined in solution of the ME and the phenomenological rate constants of a combustion model.

There remain two major areas of uncertainty in these applications.

There is still a poor understanding of the efficiency of energy transfer and of the dependence of α on the nature of both the reactive species and the collision partner and on the temperature. Direct experiments are now becoming feasible and theory

has made significant strides, but it is fair to say that a general understanding of the processes involved, which can guide the choice of α for a given system and of its T dependence, is not yet available (24, 25, 72–74).

This review has deliberately avoided any discussion of the role of angular momentum. Rotation is included in the densities of state calculations, but has not been explicitly discussed in relation to microcanonical rate constants for dissociation and isomerization. Discussion has centred on $k(E)$ and not on $k(E, J)$. There are two ways in which angular momentum enters into such calculations. Angular momentum is conserved as the molecule moves over the transition state, but because the geometry of the molecule is changing, its rotational energy is not constant. A familiar example is the centrifugal barrier that arises in radical radical reactions with no electronic potential energy barrier. A second issue relates to the role of K , the projection of J along the main symmetry axis for symmetric top molecules. If the rotations are strongly coupled to the vibrations, e.g., via coriolis interactions, then K is termed active and is not conserved. In the absence of such coupling, K is inactive and is conserved. There are well-developed techniques available for incorporating rotational effects into the calculation of microcanonical rate constants (4). The difficulty arises in incorporating these $k(E, J)$ values into the ME. The ME then becomes two dimensional, as energy transfer can involve both rotation and vibration and is much more difficult to solve. Various approaches have been devised, but there is, as yet, no agreement on the most computationally efficient, yet realistic way forward (75–78).

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